

Day 6: Advanced Power Query Transformations

📌 **Definition to Remember**

Advanced Power Query transformations allow reshaping wide datasets into normalized tall formats (**Unpivot**), aggregating tables (**Group By**), combining row sets (**Append**), and performing relational joins (**Merge**).

1. The Power of Unpivot Columns

- **Problem:** Financial tables often store months as separate columns (**Jan** , **Feb** , **Mar** ...).
- **Solution:** Select ID columns and click **Unpivot Other Columns**.
- **Result:** Transforms wide messy spreadsheets into normalized tall format: **[Attribute: Month]** and **[Value: Amount]** .

```
WIDE FORMAT (Bad for BI):
Product | Jan_Sales | Feb_Sales | Mar_Sales
Widget  | 100          | 150          | 200

TALL FORMAT (Ideal for Data Modeling):
Product | Month | Sales
Widget  | Jan   | 100
Widget  | Feb   | 150
Widget  | Mar   | 200
```

2. Append vs Merge Queries

Transformation	Type	SQL Equivalent	Description
Append Queries	Vertical Stacking	UNION ALL	Combines rows of two or more tables with identical column structures (e.g., 2024 Sales + 2025 Sales).
Merge Queries	Horizontal Join	JOIN	Merges columns from two tables based on matching key columns (Left Outer, Inner, Right Outer, Full Outer, Anti Join).

3. Group By & Custom M Columns

- **Group By:** Aggregates row values (e.g., Total Sales per Customer) directly during ETL.
- **Custom Column:** Adding columns using custom M functions (e.g., **[UnitPrice] * [Quantity] * (1 - [Discount])**).

★ Must-Master Skills & Interview Points

1. Why **Unpivot** is essential for dynamic time intelligence and DAX calculations.
2. Master the 6 Merge Join types in Power Query (especially Left Anti Join for finding orphan records).
3. When to aggregate in Power Query (Group By) vs calculating dynamically in DAX.

⚡ Quick Recall

Unpivot (Wide to Tall) + Append (Stack Rows / UNION) + Merge (Join Columns / JOIN) + Group By (ETL Aggregation)